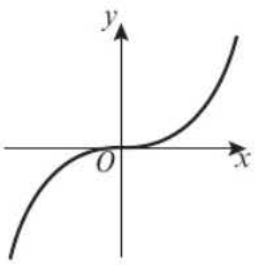
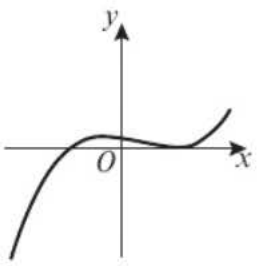
4.1 Cubic graphs

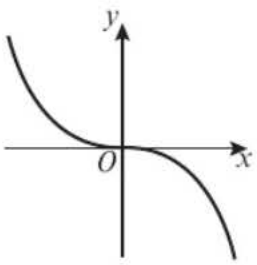
A **cubic function** has the form $f(x) = ax^3 + bx^2 + cx + d$, where a, b, c and d are real numbers and a is non-zero.

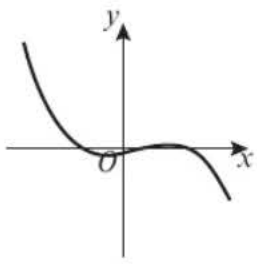
The graph of a cubic function can take several different forms, depending on the exact nature of the function.





For these two functions a is positive.





For these two functions a is negative.

- **If p is a root of the function $f(x)$, then the graph of $y = f(x)$ touches or crosses the x-axis at the point $(p, 0)$.**

You can sketch the graph of a cubic function by finding the roots of the function.

Example 1

Sketch the curves with the following equations and show the points where they cross the coordinate axes.

a $y = (x - 2)(1 - x)(1 + x)$ **b** $y = x(x + 1)(x + 2)$

a $y = (x - 2)(1 - x)(1 + x)$

$0 = (x - 2)(1 - x)(1 + x)$

So $x = 2$, $x = 1$ or $x = -1$

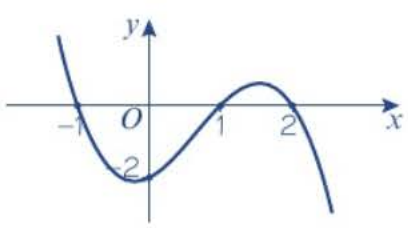
So the curve crosses the x-axis at (2, 0), (1, 0) and (−1, 0).

When $x = 0$, $y = -2 \times 1 \times 1 = -2$

So the curve crosses the y-axis at (0, −2).

$x \to \infty$, $y \to -\infty$

$x \to -\infty$, $y \to \infty$



b $y = x(x + 1)(x + 2)$

$0 = x(x + 1)(x + 2)$

So $x = 0$, $x = -1$ or $x = -2$

Put $y = 0$ and solve for x.

Find the value of y when $x = 0$.

Check what happens to y for large positive and negative values of x.

The x^3 term in the expanded function would be $x \times (-x) \times x = -x^3$ so the curve has a negative x^3 coefficient.

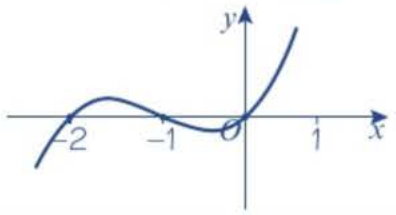
Put $y = 0$ and solve for x.

Online Explore the graph of $y = (x - p)(x - q)(x - r)$ where p, q and r are constants using GeoGebra.

So the curve crosses the x-axis at (0, 0), (–1, 0) and (–2, 0).

$x \to \infty, y \to \infty$

$x \to -\infty, y \to -\infty$



You know that the curve crosses the x-axis at (0, 0) so you don't need to calculate the y-intercept separately.

Check what happens to y for large positive and negative values of x.

The x^3 term in the expanded function would be $x \times x \times x = x^3$ so the curve has a positive x^3 coefficient.

Example 2

Sketch the following curves.

a $y = (x - 1)^2(x + 1)$ **b** $y = x^3 - 2x^2 - 3x$ **c** $y = (x - 2)^3$

a $y = (x - 1)^2(x + 1)$

$0 = (x - 1)^2(x + 1)$

So $x = 1$ or $x = -1$

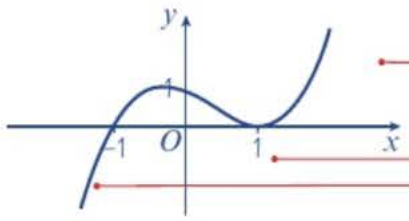
So the curve crosses the x-axis at (–1, 0) and touches the x-axis at (1, 0).

When $x = 0$, $y = (-1)^2 \times 1 = 1$

So the curve crosses the y-axis at (0, 1).

$x \to \infty, y \to \infty$

$x \to -\infty, y \to -\infty$



Put $y = 0$ and solve for x.

$(x - 1)$ is squared so $x = 1$ is a 'double' repeated root. This means that the curve just touches the x-axis at (1, 0).

Find the value of y when $x = 0$.

Check what happens to y for large positive and negative values of x.

$x \to \infty, y \to \infty$

$x = 1$ is a 'double' repeated root.

$x \to -\infty, y \to -\infty$

b $y = x^3 - 2x^3 - 3x$

$= x(x^2 - 2x - 3)$

$= x(x - 3)(x + 1)$

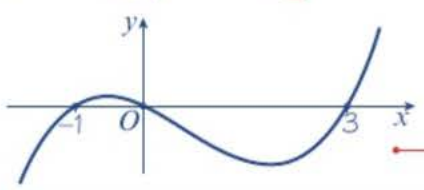
$0 = x(x - 3)(x + 1)$

So $x = 0$, $x = 3$ or $x = -1$

So the curve crosses the x-axis at (0, 0), (3, 0) and (–1, 0).

$x \to \infty, y \to \infty$

$x \to -\infty, y \to -\infty$



First factorise.

Check what happens to y for large positive and negative values of x.

This is a cubic curve with a positive coefficient of x^3 and three distinct roots.

c $y = (x - 2)^3$

$0 = (x - 2)^3$

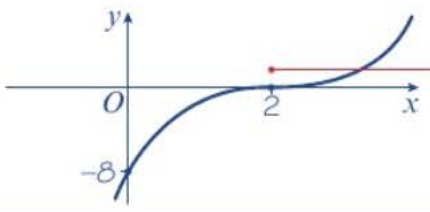
So $x = 2$ and the curve crosses the x-axis at $(2, 0)$ only.

When $x = 0$, $y = (-2)^3 = -8$

So the curve crosses the y-axis at $(0, -8)$.

$x \to \infty, y \to \infty$

$x \to -\infty, y \to -\infty$



Check what happens to y for large positive and negative values of x.

$x = 2$ is a 'triple' repeated root.

Example 3

Sketch the curve with equation $y = (x - 1)(x^2 + x + 2)$.

$y = (x - 1)(x^2 + x + 2)$

$0 = (x - 1)(x^2 + x + 2)$

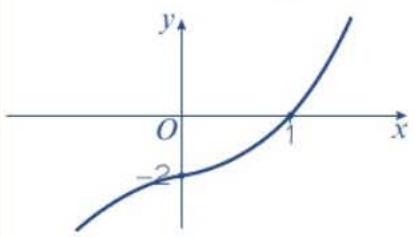
So $x = 1$ only and the curve crosses the x-axis at $(1, 0)$.

When $x = 0$, $y = (-1)(2) = -2$

So the curve crosses the y-axis at $(0, -2)$.

$x \to \infty, y \to \infty$

$x \to -\infty, y \to -\infty$



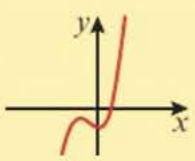
The quadratic factor $x^2 + x + 2$ gives no solutions since the discriminant $b^2 - 4ac = (1)^2 - 4(1)(2) = -7$.

← Section 2.5

Watch out A cubic graph could intersect the x-axis at 1, 2 or 3 points.

Check what happens to y for large positive and negative values of x.

You haven't got enough information to know the exact shape of the graph. It could also be shaped like this:



Exercise 4A

1 Sketch the following curves and indicate clearly the points of intersection with the axes:

a $y = (x - 3)(x - 2)(x + 1)$ **b** $y = (x - 1)(x + 2)(x + 3)$

c $y = (x + 1)(x + 2)(x + 3)$ **d** $y = (x + 1)(1 - x)(x + 3)$

e $y = (x - 2)(x - 3)(4 - x)$ **f** $y = x(x - 2)(x + 1)$

g $y = x(x + 1)(x - 1)$ **h** $y = x(x + 1)(1 - x)$

i $y = (x - 2)(2x - 1)(2x + 1)$ **j** $y = x(2x - 1)(x + 3)$

2 Sketch the curves with the following equations:

a $y = (x + 1)^2(x - 1)$ **b** $y = (x + 2)(x - 1)^2$ **c** $y = (2 - x)(x + 1)^2$

d $y = (x - 2)(x + 1)^2$ **e** $y = x^2(x + 2)$ **f** $y = (x - 1)^2x$

g $y = (1 - x)^2(3 + x)$ **h** $y = (x - 1)^2(3 - x)$ **i** $y = x^2(2 - x)$

j $y = x^2(x - 2)$

3 Factorise the following equations and then sketch the curves:

a $y = x^3 + x^2 - 2x$ **b** $y = x^3 + 5x^2 + 4x$ **c** $y = x^3 + 2x^2 + x$

d $y = 3x + 2x^2 - x^3$ **e** $y = x^3 - x^2$ **f** $y = x - x^3$

g $y = 12x^3 - 3x$ **h** $y = x^3 - x^2 - 2x$ **i** $y = x^3 - 9x$

j $y = x^3 - 9x^2$

4 Sketch the following curves and indicate the coordinates of the points where the curves cross the axes:

a $y = (x - 2)^3$ **b** $y = (2 - x)^3$ **c** $y = (x - 1)^3$ **d** $y = (x + 2)^3$

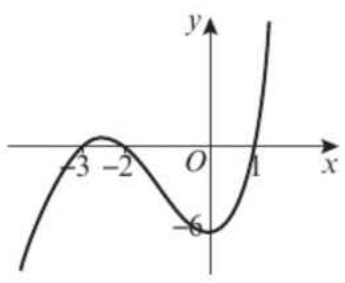
e $y = -(x + 2)^3$ **f** $y = (x + 3)^3$ **g** $y = (x - 3)^3$ **h** $y = (1 - x)^3$

i $y = -(x - 2)^3$ **j** $y = -\left(x - \frac{1}{2}\right)^3$

(P) **5** The graph of $y = x^3 + bx^2 + cx + d$ is shown opposite, where b, c and d are real constants.

a Find the values of b, c and d. **(3 marks)**

b Write down the coordinates of the point where the curve crosses the y-axis. **(1 mark)**

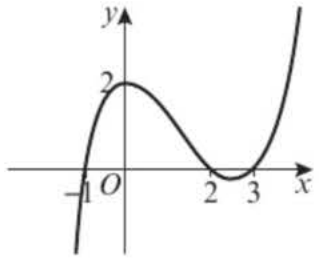


Problem-solving

Start by writing the equation in the form $y = (x - p)(x - q)(x - r)$.

(P) **6** The graph of $y = ax^3 + bx^2 + cx + d$ is shown opposite, where a, b, c and d are real constants.

Find the values of a, b, c and d. **(4 marks)**



(E) **7** Given that $f(x) = (x - 10)(x^2 - 2x) + 12x$

a Express $f(x)$ in the form $x(ax^2 + bx + c)$ where a, b and c are real constants. **(3 marks)**

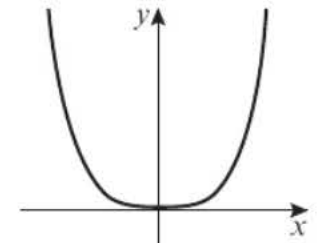
b Hence factorise $f(x)$ completely. **(2 marks)**

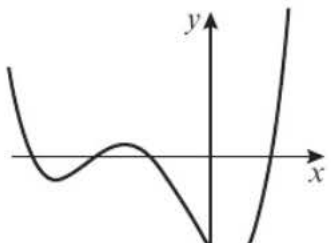
c Sketch the graph of $y = f(x)$ showing clearly the points where the graph intersects the axes. **(3 marks)**

4.2 Quartic graphs

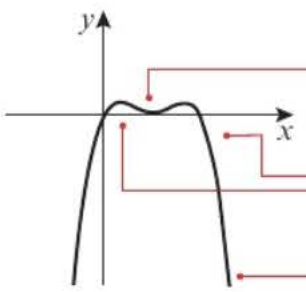
A **quartic function** has the form $f(x) = ax^4 + bx^3 + cx^2 + dx + e$, where a, b, c, d and e are real numbers and a is non-zero.

The graph of a quartic function can take several different forms, depending on the exact nature of the function.





For these two functions a is positive.



This is a repeated root.

These roots are distinct.

For this function a is negative.

You can sketch the graph of a quartic function by finding the roots of the function.

Example 4

Sketch the following curves:

a $y = (x + 1)(x + 2)(x - 1)(x - 2)$ **b** $y = x(x + 2)^2(3 - x)$ **c** $y = (x - 1)^2(x - 3)^2$

a $y = (x + 1)(x + 2)(x - 1)(x - 2)$

$0 = (x + 1)(x + 2)(x - 1)(x - 2)$

So $x = -1, -2, 1$ or 2

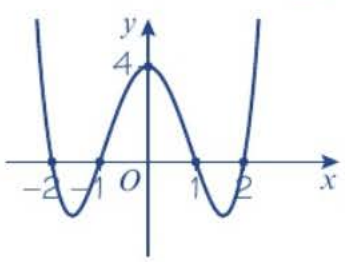
The curve cuts the x-axis at $(-2, 0)$, $(-1, 0)$, $(1, 0)$ and $(2, 0)$.

When $x = 0$, $y = 1 \times 2 \times (-1) \times (-2) = 4$.

So the curve cuts the y-axis at $(0, 4)$.

$x \to \infty, y \to \infty$

$x \to -\infty, y \to \infty$



Set $y = 0$ and solve to find the roots of the function.

Substitute $x = 0$ into the function to find the coordinates of the y-intercept.

Check what happens to y for large positive and negative values of x.

We know the general shape of the quartic graph so we can draw a smooth curve through the points.

Online Explore the graph of $y = (x - p)(x - q)(x - r)(x - s)$ where p, q, r and s are constants using GeoGebra.